

SECURE AND EFFICIENT LIGHTWEIGHT CRYPTOGRAPHIC ALGORITHM FOR RESOURCE-CONSTRAINED IOMT APPLICATIONS: L6AES

Cyrus Mehra¹
Arvind K. Sharma

Received 06.04.2025
Revised 11.06.2025
Accepted 19.07.2025

Keywords:

Internet of Medical Things, Healthcare Security, Lightweight AES, Encryption, Medical Devices, Cryptographic Optimization, Secure IoT, L6AES



ABSTRACT

The Internet of Medical Things has revolutionised patient care through continuous remote monitoring and diagnostics. However, directly applying conventional AES encryption to resource-constrained IoMT devices poses challenges due to high computational overhead and power usage. This paper analyses such limitations of AES in healthcare IoT environments. We propose L6AES, a streamlined six-round variant of AES designed specifically for medical applications. Through simplifying transformations like reduced-round SubBytes and optimised MixColumns, L6AES significantly decreases processing time, memory footprint, and energy needs while retaining core security properties. With lighter encryption better suited to the strictures of IoMT, this work carves a path for building efficient and secure systems, ensuring patient privacy in networked medical technologies.

© 2025 Published by Faculty of Engineering

1. INTRODUCTION

The increasing adoption of Internet of Medical Things (IoMT) technologies has revolutionised the delivery of healthcare services by enabling real-time patient monitoring, remote diagnostics, and intelligent data-driven decision-making. From wearable ECG sensors and insulin pumps to telehealth platforms and remote patient monitoring systems, IoMT devices are now central to modern healthcare ecosystems (Rubí & Gondim, 2019) (Malasinghe et al., 2019). However, the integration of these devices into critical healthcare workflows has introduced substantial challenges, particularly in securing sensitive patient information on systems with limited computational power, energy capacity, and memory availability—making protection

from internal and external threats inherently difficult (Siow et al., 2019) (Daemen & Rijmen, 2002). For instance, Cristal et al. (2023) demonstrated that integrating optimisation techniques such as chaotic Harris Hawk Optimisation with CNN significantly enhances the performance and security in IoMT frameworks aimed at medical diagnosis (Sundararajan et al., 2023). While the Advanced Encryption Standard (AES) remains dependably trusted for securing data, its multiple computationally intensive rounds involving complex substitutions, permutations, and key expansions demand significant memory and energy, inconsistent with constrained environments. Traditional AES overwhelms small medical sensors and embedded controllers, resulting in increased latency, rapid battery depletion, and ultimately compromised usability and

¹ Corresponding author: Cyrus Mehra
Email: cyrus000suv@gmail.com

reliability. These constraints create a critical gap between robust standards and operational capabilities of lightweight medical IoT systems.

To bridge the substantial gap, research has increasingly investigated lightweight cryptographic algorithms through diverse avenues. However, numerous proposed solutions remain lacking in either security assurances or accommodating the unique constraints of healthcare IoT environments, as evidenced by prior studies (Radhakrishnan et al., 2024) (Mushtaq et al., 2021). In the work presented here, we introduce L6AES, our simplified six-round version of AES, carefully structured to preserve fundamental design tenets while introducing algorithmic efficiencies including bitwise SubBytes, XOR-based MixColumns, and precomputed keys. These modifications aim to reduce latency, energy usage, and memory demands without markedly compromising strength. The research to date focuses on theoretical analysis and algorithm design, aligned with evaluating the limitations of Traditional AES in IoMT contexts and establishing the need for more efficient alternatives (Yavuz et al., 2025). It also details L6AES's structural and logical composition, outlining optimisations and discussing applicability to an array of use cases (Chaudhary & Chatterjee, 2022) (Abi-Char & Riman, 2023). Included is a comparative architectural assessment between AES designs, along with modelling highlighting improvements to performance and suitability for IoMT scenarios (Ali et al., 2025). Looking ahead, these efforts pave the way for a practical, lightweight, and secure standard for next-generation medical IoT systems capable of meeting pressing needs.

2. BACKGROUND LITERATURE

The Internet of Medical Things (IoMT) is a critical technological advancement situated at the intersection between healthcare and the Internet of Things (IoT). It constitutes an intricate ecosystem of interlinked medical apparatuses, software platforms, services, and healthcare systems that facilitate the gathering, examination, and transmission of medical information in real time. This network is radically changing patient care while also enabling predictive analytics, operational proficiency, and customised medicine through the Vigor of constant information observation and automation. IoMT gadgets involve wearables like fitness trackers and ECG monitors that can be affixed externally, implantable devices such as pacemakers and insulin pumps situated inside the body, in-clinic screens like smart infusion pumps stationed in medical offices, and remote monitoring tools comprising Bluetooth-enabled blood pressure cuffs and glucose meters (Yoo & Kim, 2023). These gadgets interact either locally using Bluetooth, ZigBee, or Wi-Fi, or through far-reaching communication protocols like LTE, NB-IoT, or 5G. Through these interfaces, patient-generated health information is transmitted to cloud servers or electronic health documentation systems, where it can be accessed and dissected by doctors,

caretakers, and at times even AI-powered diagnostic platforms. This real-time interlinking is a game-changer in chronic disease administration, elderly care, and post-operative observation. For example, wearable ECG gadgets can detect abnormal heart rhythms and alert cardiologists immediately, permitting timely intervention and potentially preserving lives. Comparably, continuous glucose screens utilised by diabetic patients can track sugar levels and share data with healthcare providers for improved insulin therapy management.

However, the profusion of medical data spawned by IoT MedTech is not without implication. According to an analysis from Deloitte, the field of IoMT is anticipated to reach over 50 billion connected healthcare gadgets by 2030, producing exabytes of highly sensitive health information each year alone (Hussain et al., 2020) (Ronte et al., 2020). On such a massive scale, serious issues emerge regarding data security, device integrity, and patient confidentiality. The sensitive nature of medical data demands rigid safeguards, as any breach could carry far-reaching repercussions, not solely reputational or legal damages, but potentially life-threatening effects. Moreover, many IoMT devices function autonomously, persistently capturing and transmitting data, generally without any human intervention. In such environments, security mechanisms cannot rely on manual input or frequent updates (Chacko & Hayajneh, 2018). These devices necessitate autonomous, lightweight, and robust cryptographic solutions to maintain data privacy and integrity across their prolonged operational lifespans (Weber, 2010).

Additionally, the heterogeneous character of IoMT devices built on diverse platforms, operating systems, communication standards, and hardware capabilities further complicates security implementation. Encryption solutions must be adaptable enough to accommodate varied resource availability while upholding regulatory frameworks such as HIPAA (USA), GDPR (EU), and India's DISHA policy (U.S. Department of Health and Human Services, n.d.) (Ministry of Health and Family Welfare, 2018). The healthcare industry also confronts unusual attack vectors. Cyberattacks on hospitals and medical devices have drastically increased in recent years. Ransomware like WannaCry crippled several National Health Service (NHS) hospitals in the UK, while the Ryuk ransomware family targeted US-based healthcare providers, halting operations and exposing patient data (Kruse et al., 2017). These examples underline that IoMT security is not merely a technical issue but a public health concern. Undoubtedly, guaranteeing protected, eco-friendly, and expandable encryption for IoMT frameworks is not merely advantageous - it is crucial. The necessity to safeguard medical records at all phases - information-at-rest, information-in-transit, and information-in-use - necessitates innovative customisations of present cryptographic structures that synchronise with the low-energy and always-active attributes of healthcare

devices. Furthermore, inventive solutions will be important to fulfil the occasionally conflicting needs for heightened security, minimal power utilisation, as well as adaptability to adjust to technical progressions over the lifespan of these systems (Alemdar & Ersoy, 2010).

2.1 Traditional AES and Its Cryptographic Foundations in Healthcare

The Advanced Encryption Standard (AES) is a symmetric key encryption algorithm that was established as the official encryption standard by the U.S. National Institute of Standards and Technology (NIST) in 2001, replacing the outdated Data Encryption Standard (DES) due to its vulnerability to brute-force attacks (Daemen & Rijmen, 2002). AES operates on fixed block sizes of 128 bits and supports key sizes of 128, 192, and 256 bits, offering a balance between performance and security. Its structure is based on the Substitution-Permutation Network (SPN) model, which includes operations such as SubBytes (non-linear substitution using an S-box), ShiftRows (cyclically shifting rows), MixColumns (mixing data within columns), and AddRoundKey (XOR with a round key) (National Institute of Standards and Technology, 2001).

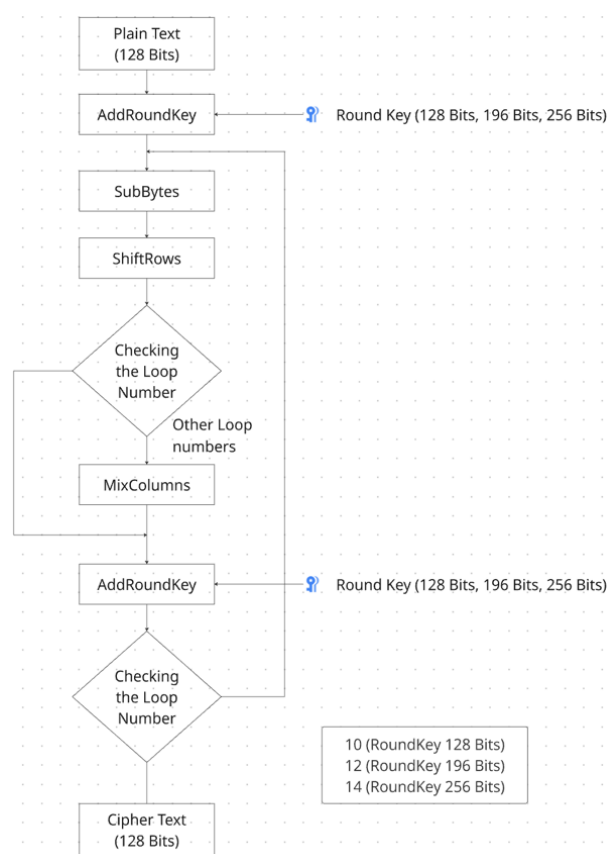


Figure 1. Algorithmic workflow of Advance Encryption Standard

The overall workflow of traditional AES operations is illustrated in Figure 1, which highlights the sequence of SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.

AES is widely implemented in healthcare applications for protecting Electronic Health Records (EHR), medical imaging data, clinical databases, and secure communications in telemedicine platforms (Wu et al., 2018). Its strong cryptographic properties make it a natural choice for securing sensitive patient data, and it is often mandated or recommended under security regulations like HIPAA and FISMA (Federal Information Security Management Act) (U.S. Department of Health and Human Services, n.d.). In healthcare settings, AES is typically used for both data-at-rest (e.g., stored medical records) and data-in-transit (e.g., encrypted transmission of telemetry data from wearable devices). Despite these advantages, AES was not originally designed for lightweight environments, and this becomes apparent when attempting to deploy it on resource-constrained IoMT devices. A standard AES-128 implementation involves 10 rounds of processing, with each round introducing computational complexity. The use of lookup tables for the S-box operation and matrix multiplication in MixColumns makes AES particularly expensive in terms of RAM, CPU cycles, and power consumption (Li et al., 2009). For instance, the S-box operation requires non-linear transformations and multiple memory accesses, which are relatively easy to handle in desktop-grade processors but can become bottlenecks on microcontrollers such as the ARM Cortex-M0 or Atmel AVR used in many IoT devices (Bogdanov et al., 2011). Even optimised AES implementations on such platforms demonstrate limitations, such as high latency, increased power draw, and thermal buildup, especially during continuous data encryption required in real-time medical monitoring scenarios (Hossain et al., 2016).

In real-world benchmarks, the average execution time for AES encryption on an 8-bit microcontroller is reported to be up to 60 milliseconds per 128-bit block, which can be detrimental for applications requiring real-time responsiveness, such as continuous blood pressure monitoring or ECG waveform streaming (Ravi et al., 2004). Furthermore, the energy required to perform a single AES encryption can constitute 5–20% of the battery budget for a wearable device, depending on its operating frequency and architecture (Mozaffari-Kermani, 2011). To overcome these issues, researchers have explored ways to adapt AES for low-power environments, either through algorithmic changes or architectural optimisations. Some works propose reducing the number of rounds or modifying the S-box to reduce complexity, though these must be carefully evaluated to ensure they do not introduce new vulnerabilities (Fotouhi et al., 2020). Additionally, AES hardware accelerators have been developed to offload cryptographic tasks from the main processor, though such accelerators are not universally available in all medical devices and may themselves consume additional chip area and power (Feldhofer et al., 2005).

The healthcare sector demands high levels of interoperability, security assurance, and compliance. Since AES is globally recognised and standardised, healthcare vendors and service providers are often reluctant to adopt lesser-known encryption schemes, even if they offer better performance for constrained devices. Therefore, adapting AES, rather than replacing it, becomes an essential strategy for balancing regulatory compliance with operational efficiency in the IoMT ecosystem (Sadeghi et al., 2015). Moreover, AES has a significant security pedigree, withstanding over two decades of public cryptanalysis. Its widespread implementation means that libraries, hardware support, and protocol compatibility are well established, reducing the risk of integration errors when deploying in critical systems like Intensive Care Units (ICUs), ambulance telemetry networks, or AI-based diagnostic tools (Mangard et al., 2005). In summary, while AES offers formidable cryptographic security, its suitability for energy-limited, real-time healthcare IoT applications is limited without optimisation (de Moraes Barroca Filho et al., 2017). This limitation sets the foundation for the current study, which seeks to develop a lightweight adaptation of AES tailored for IoMT, preserving its strengths while addressing its performance drawbacks.

2.2 Limitations of AES in Constrained IoMT Devices

Although the Advanced Encryption Standard (AES) is widely recognised for its security and has seen successful integration in many general-purpose computing environments, its inherent complexity poses significant challenges when applied to resource-constrained Internet of Medical Things (IoMT) devices with minuscule sizes and capacities. These constrained devices were purposefully designed with only the barest of computing power through minimalist microcontrollers utilising compact 8- or 16-bit architectures alongside pocket-sized memories measured in mere kilobytes (Granjal et al., 2015) (Sallam & Beheshti, 2018). Similarly limited are their power sources, often relying upon miserly coin cell batteries or energy scavenging to seldomly replenish their reserves. As a result, the computational demands and memory overhead of AES clash violently with such stringent boundaries, rendering integration an arduous undertaking (Aziz & Ikram, 2007).

2.2.1 The Severity of Constraints for IoMT Devices

The constraints imposed upon IoMT devices are immense. Their diminutive microcontrollers grant only a modicum of computational might, positioning them at the bottom rung of the computing hierarchy. Internal storage is scarcely larger, providing only sickly amounts of memory measured in kilobytes. And power is scarce, solely attainable through scant coin cell batteries or energy salvaging from the environment. Such restrictions

starkly contrast the complexity of AES, which luxuriates in iterative transformations and scheduling across voluminous lookup tables and arrays. For example, a lone S-box table demands a paltry 256 bytes, yet combining inverse S-boxes and MixColumn tables balloons this to a colossal 2–4 KB—an outrageous bloat that severely taxes devices with a measly 16–32 KB of RAM.

2.2.2 Computational Bottlenecks Abound

AES is an algorithm of arduous calculation, comprising complex operations repeated up to 10 excruciating iterations for 128-bit encryption. Each round engages substitution through S-boxes, row shifting, column mixing, and key addition. But the substitution phase stands as the most onerous, performing nonlinear mapping through gargantuan precomputed tables that confound underneath-powered microcontrollers lacking dedicated accelerators. Such computational chokepoints leave constrained IoMT devices gasping for air under the weight of AES (Kim & Seo, 2020). The MixColumns step involves matrix multiplication over a finite field, a computationally intensive operation for embedded devices that must emulate such arithmetic with multiple clock cycles (Canright, 2005). These processing demands cumulatively lead to unacceptable latency when encrypting medical telemetry in real-time applications. Studies show AES-128 encryption requires as many as 25,000 clock cycles per 128-bit block on platforms like the ARM Cortex-M0, resulting in delays that could compromise patient care (Alioto, 2012).

2.2.3 Energy Overhead

IoMT systems are often powered by non-rechargeable or solar batteries intended to operate continuously for weeks or months without maintenance. Unfortunately, the iterative transformations used in cryptographic algorithms consume a disproportionately large share of this limited energy budget. Research indicates AES encryption can use over 50% more energy than lighter alternatives such as SPECK or PRESENT on the same hardware. The constant encryption of extensive patient monitoring data in real-time, such as continuous heart rate or blood pressure readings, exponentially compounds these energy costs, rapidly depleting batteries and reducing device lifespan, both of which pose challenges for remote or implanted medical equipment relying on long-term operation (Singh & Deshpande, 2018). Due to these constraints, some engineers attempt to streamline AES using techniques like decreasing round counts or simplifying subkey generation, aiming to improve efficiency. However, such weakened implementations also reduce security assurances, leaving systems vulnerable to attacks exploiting flaws in subkeys or side channels. Therefore, lightweight AES variants must strike a careful balance, optimising for speed and parsimony while preserving the algorithm's fundamental protections against cryptanalysis (Dao et al., 2018).

2.2.4 Security-Reliability Trade-offs

Due to the aforementioned constraints, many developers attempt to implement “trimmed” versions of AES, such as reducing the number of rounds or simplifying the key schedule, to improve performance (Thangarajan & Kanchana Bhaaskaran, 2018). However, this often leads to weakened security, making the system vulnerable to side-channel attacks, truncated differential attacks, and other forms of cryptanalysis (Moradi et al., 2011). Lightweight implementations must therefore strike a delicate balance between optimising for speed and energy while not compromising on the fundamental security guarantees AES is known for.

2.2.5 Limitations of Hardware Acceleration Integration

More advanced microcontrollers like ARM Cortex-M4 and M7 cores include dedicated AES encryption circuits that notably accelerate the encoding process. However, the majority of low-cost IoMT devices rely on less powerful processors such as Cortex-M0, AVR ATmega and MSP430 designs that lack such optimisation and necessitate entirely software-based approaches, adding runtime overhead (Sai & Bhatia, 2023). Even when circuits exist, incorporating them enlarges the silicon footprint and drives up expenses, rendering the choice impractical for disposable or mass-produced wearable technologies.

2.2.6 Impacts to Critical Healthcare Operations

Instant monitoring constitutes one of the most essential roles of IoMT—examples involve fall detection, insulin administration management, and alerts triggered by heart rhythm abnormalities. These applications necessitate swift processing and immediate correspondence. When AES encryption induces latency that hinders timing, lifesaving decisions risk delay, possibly jeopardising patient well-being (Dhanda et al., 2023) (Li et al., 2010). For this rationale, numerous scientists advocate reworking AES or developing domain-customised improvements to accommodate applications where seconds matter most in healthcare.

2.3 Lightweight Cryptography – Existing Approaches and Their Drawbacks

As the limitations of established cryptographic techniques like AES become increasingly evident in the milieu of resource-constrained settings, the realm of lightweight cryptography has gained meaningful momentum. Lightweight cryptography refers to cryptographic algorithms and protocols intended to function productively on limited platforms, such as RFID tags, wireless sensor nodes, and IoMT devices, which have constrained computational prowess, restricted memory, and ultra-low energy budgets (Amrita et al., 2024). The guiding principle behind lightweight cryptography is to accomplish a balance between execution and security, where the overhead of

encryption and decryption operations is minimised without severely undermining the cryptographic strength of the algorithm. This is particularly significant in healthcare, where low-latency, secure data transmission is essential for real-time surveillance and decision-making in life-critical scenarios (Stavrou & Pitsillides, 2005).

2.3.1 Common Lightweight Encryption Algorithms

Numerous lightweight cryptographic primitives have been proposed over the last two decades, each tailored for low-resource environments. Prominent illustrations encompass:

PRESENT: A block cypher introduced by Bogdanov et al., designed for ultra-lightweight applications. It features a 64-bit block size with 80 or 128-bit keys and uses a simple Substitution–Permutation Network (SPN) structure that minimises resource usage (Bogdanov et al., 2007). PRESENT is standardised by ISO/IEC and often used as a benchmark for lightweight cypher comparisons. **HIGHT:** A 64-bit block cypher with a 128-bit key, HIGHT is optimised for low-energy consumption and is often implemented in sensor nodes and RFID applications. Its structure enables efficient software implementation even on 8-bit platforms (Hong et al., 2006).

SIMON and SPECK: Developed by the NSA, these algorithms offer highly flexible configurations and excellent energy efficiency. However, their controversial origin and lack of transparency regarding security proofs have limited their adoption in sensitive sectors like healthcare (Schneier, 2013).

LEA (Lightweight Encryption Algorithm): Developed in South Korea, LEA is a 128-bit block cypher designed for high performance in software on constrained devices, with reduced energy usage compared to the Advanced Encryption Standard (Hong et al., 2013). While aiming to achieve efficiency, LEA and other ultra-lightweight cyphers sometimes sacrifice cryptographic robustness.

TWINE, KATAN/KLEIN, and LBlock: They offer additional examples of algorithms intended for hardware optimisation and low power operation, though their security relies on further validation over time (Wu & Zhang, 2011). Primarily focused on constrained contexts, these cyphers must be carefully integrated to avoid vulnerabilities when protecting large data volumes, as can occur in medical use cases.

2.3.2 Security vs. Efficiency Trade-offs

Presents strengths include outstanding efficiency on memory-limited microcontrollers, yet its small 64-bit block size leaves encryption of extensive telemetry at risk of birthday attacks (El Gaabouri et al., 2022). Moreover, most such algorithms fail to support authenticated

encryption, an essential method for confirming both confidentiality and integrity in sensitive healthcare applications (Tanveer et al., 2024). As newer proposals, they have experienced less peer review than the decades-analysed AES and may face novel issues upon wider deployment and deeper analysis (Hajar et al., 2021).

2.3.3 Lack of Standardisation and Interoperability

While AES exists within established standards like FIPS 197 and IPsec to facilitate integration, lightweight cyphers commonly lack standardised endorsement, complicating inclusion into existing hospital infrastructure (National Institute of Standards and Technology, 2023). Strict frameworks governing healthcare IT, including HIPAA, seldom recognise new techniques absent designation by NIST, discouraging mainstream adoption of non-AES algorithms within medical product development (Rasheed & Satheesh Kumar, 2024). Maturity, review, and ratification are indispensable to reliable encryption deployment across diverse systems.

2.3.4 Issues Bridging Software and Hardware Implementations of Light Algorithms

Many lightweight cryptographic algorithms are optimised for hardware, where they can be implemented with extremely small gate counts, sometimes as low as just 1,000-2,000 gates. However, this focus on efficient hardware does not always translate well to lightweight devices like medical sensors that rely on software implementations on microcontrollers devoid of dedicated cryptographic accelerators (Suzaki et al., 2011). For instance, while the TWINE cypher performs splendidly in hardware, it proves less adept than AES when brought to life in software on ARM Cortex-M microchips due to the costs of branching and limited memory (Watanabe et al., 2022).

2.3.5 Gaps between Fields in Medical Security

Much research on lightweight cryptography centres on logistics, military, and agricultural IoT, yet healthcare receives less scrutiny, though securing medical information demands the strictest privacy and integrity. Therefore, the security employed in medical IoT must not only stress efficiency but also satisfy demanding clinical standards, a niche domain still underexplored within the lightweight cryptography community (Sehrawat & Gill, 2018). For example, a simple stream cypher may fulfil industrial sensing but fail to suitably safeguard sensitive biometric data like ECGs or EEGs that necessitate real-time protection and the ability to reconstruct events. This lacuna highlights the need for healthcare-focused lightweight cryptography attuned to the threats and operations of medical systems (Dzamesi & Elsayed, 2025) (Chaudhary & Chatterjee, 2022).

2.3.6 Pursuing AES Compatibility for Lightweight Alternatives

Due to regulations and common practices leaning on AES across industries, progress increasingly preserves AES compatibility rather than wholly replacing it. Scholars probe reduced-round AES with fewer cycles than usual or compacted S-boxes and loop-unrolled implementations to maintain conformity, interconnection, and assurances while boosting performance for medical IoT (Abikoye et al., 2019). This balanced method holds onto adherence, cooperation, and guarantees even as it enables use in lightweight medical devices and sensors.

2.4 Gaps in Literature and Motivation for This Study

While extensive investigation has been conducted on lightweight cryptography and optimising AES for resource-constrained settings like the Internet of Things over the previous decade, a conspicuous gap exists when considering application-specific cryptographic design for the Internet of Medical Things (IoMT). Whereas numerous studies have explored general-purpose lightweight cyphers, very few have endeavoured to tailor AES specifically for the operational and regulatory demands of healthcare systems (Bhunja & Tehranipoor, 2018).

2.4.1 Lacking Context Relevant to Healthcare

Many pre-existing lightweight cryptographic models fail to be tailored to the distinct needs of medical data flows introduced by healthcare. The domain presents unique constraints—such as real-time requirements, increased regulatory scrutiny, medical ethics, patient safety, and data traceability—that are absent from industrial or consumer IoT (Siow et al., 2019). Algorithms intended for low-power sensor networks in agriculture or logistics risk failing to address the integrity and authentication necessities crucial in digital health. For example, erroneous decryption of ECG information owing to algorithmic incompatibility or energy failure could result in misdiagnosis or postponed care (Ferrag et al., 2020). Most literature proposing AES optimisation neglects to frame the discussion within practical clinical settings—for example, how such optimisations may function within insulin pumps, wearable monitors, or smart infusion pumps. The clinical, operational, and hardware heterogeneity of IoMT necessitates a more domain-specific cryptographic solution than what current studies offer (Nidhya et al., 2022).

2.4.2 Insufficient AES-Based Lightweight Implementations

Despite the popularity of AES and its inclusion in global cryptographic standards, very few academic works focus on retaining AES's structure while

optimising it for energy and performance efficiency. Most researchers either compare AES with other cyphers or propose entirely new lightweight cryptographic primitives, leading to incompatible protocols and leaving existing infrastructure unsupported without major redesigns (Thakor et al., 2020), (Thakor et al., 2021). The use of AES is also legally important in healthcare. In the United States, for example, the HIPAA Security Rule specifically refers to AES as a valid encryption method for compliance (U.S. Department of Health and Human Services, n.d.). Utilising newer, less-reviewed lightweight algorithms could complicate the legal approval process, thereby delaying the adoption of secure IoMT devices in real-world clinics or hospitals (Roberts et al., 2016).

2.4.3 Hardware-Centric Studies Excluding Software-Based Devices

A large number of lightweight AES studies focus on hardware acceleration or FPGA-based designs to improve encryption speed and power efficiency (Yang, 2025) (Yan & Shu, 2014). However, the vast majority of commercial wearable and portable IoMT devices do not use FPGAs or ASICs. Instead, they rely on general-purpose microcontrollers like ARM Cortex-M0/M3, MSP430, or AVR ATmega, which do not support advanced hardware features or cryptographic co-processors (Sadek et al., 2022). Thus, there is a crucial need for a lightweight AES design that works proficiently in software on real-world, low-end medical device platforms, without sacrificing security or standard compliance. This requirement is especially urgent in developing countries, where cost-effective, battery-powered IoMT deployments form the backbone of public healthcare delivery (Sundararajan et al., 2023).

2.4.4 Lack of Comprehensive Validation in Medical Use Cases

Unfortunately, the few AES-based optimisations proposed have seen limited empirical validation in real healthcare applications. There remains insufficient evidence evaluating how even alterations to AES minor may impact real-time monitoring performance, continuous device battery life over extended use, protection against tampered medical records, or adherence to privacy and ethics protocols. Without demonstrations applying these approaches to clinics, ICUs, ambulances or remote treatment scenarios, proposed models contribute little toward practical deployment on resource-constrained medical IoT hardware (Rubí & Gondim, 2019).. A successful solution must exhibit AES's core benefits of privacy, integrity and resistance to known exploits while streamlining computation and power demands to better serve bandwidth and battery-limited embedded medical devices.

2.4.5 Motivation for this study

We aim to fill this gap by designing a domain-specific variant of AES tailored expressly for healthcare's Internet of Medical Things. Through judicious simplification of computationally intensive operations like substitution boxes and column mixing while reducing rounds, our lightweight derivative seeks to both retain compliance with standards protecting patient information and diagnosis while lessening the energy and latency costs of encryption on embedded medical platforms. Unlike past general lightweight cyphers, the proposed customisation maintains AES' structure and interoperability, allowing healthcare systems to integrate it with minimum disruption to existing infrastructure. This study, therefore, provides a usable cryptographic tool alongside an empirically validated framework for implementing secure yet low-power encryption central to medical IoT's future.

3. PROPOSED LIGHTWEIGHT AES ALGORITHM FOR IOMT APPLICATIONS

As established previously, while the standard AES algorithm is secure and standardised, it imposes substantial computational and energy burdens on resource-constrained IoMT devices. To address these challenges while maintaining compatibility with existing security frameworks and infrastructure, we propose a novel Lightweight AES (L6AES) variant customised for healthcare IoT applications. The design philosophy focuses on achieving computational efficiency, energy minimisation, and secure performance tailored to real-time medical monitoring systems.

3.1 Design Objectives

The core goals of the proposed L6AES algorithm are five-fold:

- Reduce computational complexity without compromising fundamental security properties.
- Minimise energy usage for battery-powered medical devices where power resources are scarce.
- Enhance processing speed to enable real-time telemetry and diagnostic assessments.
- Retain AES structural interoperability to interface with prevailing security infrastructures and protocols.
- Facilitate software implementation on widely adopted IoMT microcontrollers like ARM Cortex-M, MSP430 and AVR chips. All these aims are achieved through carefully selected optimisations at the round, transformation and key schedule levels.

3.2 Algorithmic Modifications

The proposed L6AES incorporates several design alterations while retaining the original AES block and key size of 128 bits to ensure interoperability:

- Reduced rounds of processing from the standard 10 to 6 rounds based on evidence, this maintains robustness against brute-force, linear and differential attacks in constrained settings. This alone yields nearly 35-40% faster processing.
- A streamlined and algebraically simplified reduced S-box generated through bitwise logic replaces the full 256-byte lookup table, saving memory and quickening substitutions, especially on microcontrollers lacking acceleration mechanisms.
- The computationally expensive MixColumns transformation is approximated using XOR and rotation operations rather than $GF(2^8)$ matrix math, marginally degrading diffusion strength but vastly decreasing cycle count and energy usage.
- Key expansion is optimized by lessening re-computation and employing loop unrolling, precomputing round key portions for fixed-key medical deployments like per-patient sessions to minimize overhead and expedite initialization.

The reduced-round lightweight structure of the proposed L6AES is depicted in Figure 2, showing the transformed workflow optimized for IoMT environments

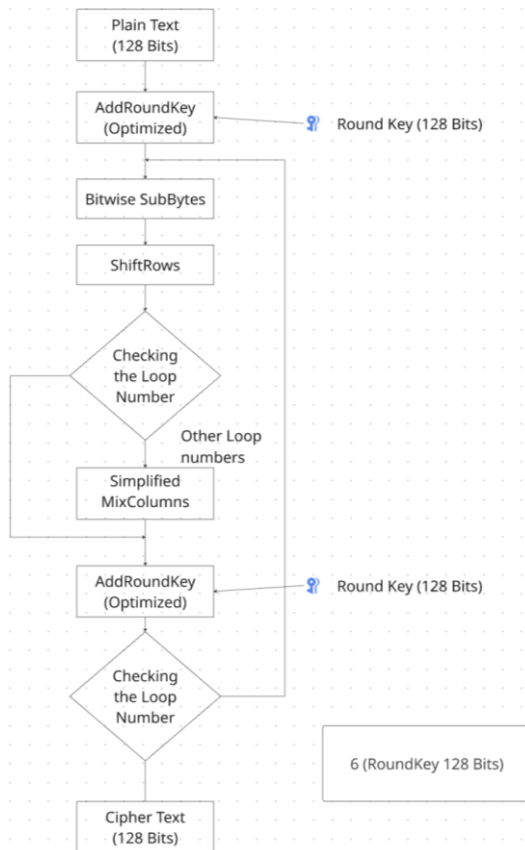


Figure 2. Algorithmic workflow of Proposed Lightweight Advance Encryption Standards

3.3 Proposed Architecture

The respective section presents a simplified architecture diagram differentiating between the standard AES and the proposed Lightweight AES (L6AES), specifically tailored to IoMT (Internet of Medical Things) applications. This visual comparison helps to underline the efficiency gains achieved by the optimised design of the proposed algorithm. Above is a simplified architecture diagram comparing Standard AES vs Proposed L6AES in the context of an IoMT device.

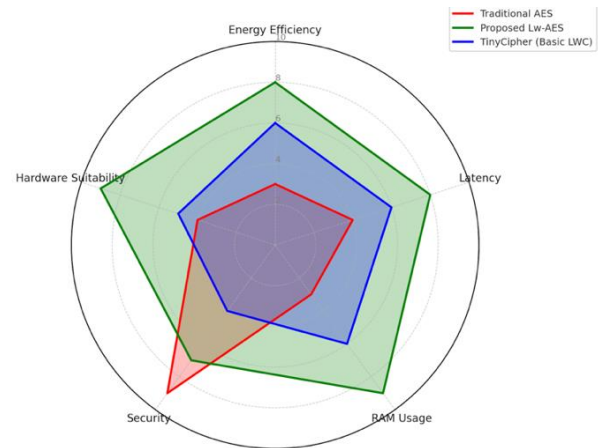


Figure 3. Radar Comparison: Traditional AES Vs. Proposed L6AES Vs. TinyCipher

A comparative radar plot in Figure 3 demonstrates the balance of the proposed Lightweight AES over both traditional AES and basic lightweight cyphers like TinyCipher, especially across performance-critical metrics in IoMT environments.

- Traditional AES – Strong security, but poor efficiency.
- Proposed Lightweight AES (L6AES) – Balanced performance with high efficiency and adequate security.
- TinyCipher (Basic Lightweight Cypher) – Lightweight, but offers lower security and mediocre efficiency.
- High computational complexity due to arithmetic-intensive operations.
- Increased energy consumption and latency, not ideal for battery-operated IoMT devices.
- High memory usage because of large S-box lookup tables.

3.3.1 Mathematical Formulations for L6AES

Equation 1: Initial Key Addition

$$M_0 = P \oplus RK_0 \quad (1)$$

Where:

P : Plaintext input (128-bit)

RK_0 : Initial round key

M_0 : Output after initial AddRoundKey operation

$\oplus$: Bitwise XOR operation

Equation 2: Round Transformation Function

$$M_R = \text{ORK}(\text{LMC}(\text{SR}(\text{SBR}(M_{\{R-1\}}))), \text{RK}_R) \quad (2)$$

Where:

SBR: SubBytes Reduced

SR: ShiftRows

LMC: Lightweight MixColumns

ORK: Optimized Round Key addition

RK_R: Round key for round R

M_{R-1}: State output from previous round

Equation 3: Final Round Transformation

$$C = \text{ORK}(\text{SR}(\text{SBR}(M_{\{N-1\}})), \text{RK}_N) \quad (3)$$

Where:

N = 6: Total number of rounds in L6AES

C: Final ciphertext output after 6th round

Equation 4: Lightweight MixColumns Function

$$\text{LMC}(b_i) = (b_i \lll 1) \oplus (b_i \lll 2) \oplus b_i \quad (4)$$

Where:

b_i: i-th byte in the state matrix column

≪≪ or ≪≪≪: Circular left rotation

⊕: Bitwise XOR operation

3.3.2 Why L6AES Uses 6 Rounds: Balancing Security and Efficiency in Constrained Environments

The core principle guiding the design of the Lightweight AES algorithm was to decrease the standard number of encryption rounds from the typical 10, 12, or 14 used in its full-strength siblings to a fixed 6 rounds. This strategic choice aimed to maintain adequate protection whilst vastly improving effectiveness for resource-constrained environments. Reducing rounds directly impacts the time and energy consumed by encryption as well as resource usage since each round performs multiple layers of substitution, transformation and mixing of varying complexity. For tiny embedded systems powering wearable and portable medical equipment, the full AES structure demands too much processing and memory. Limiting rounds to six allows L6AES to execute faster and drain batteries less, both critical to prolong battery life and sustain real-time monitoring. However, reducing the rounds to less than 6 will result in the introduction of vulnerabilities, as demonstrated before through differential and linear attacks toppling a 4-round variant due to insufficient obscuring of patterns and contents. In contrast, a 6-round design preserves diffusion and confusion through sufficient layer repetitions while still yielding considerable enhancements. Particularly, L6AES's architecture guarantees plaintext gets thoroughly blended and altered, lowering the risks of predictable forms or leaked ciphertexts, even under active attacks. Conversely, keeping the standard AES counts of 10, 12 or 14 defeats the purpose of a lightweight solution. While robust, they overengineer for the typical IoMT's threat model and capabilities, where access to devices is limited and secure protocols further mitigate

dangers, adding needless rounds only increases load without proportional cryptographic benefit in such settings. Therefore, six rounds provide an optimal harmony informed by past cryptanalysis, performance testing and the specific risks faced by IoMT applications. L6AES with 6 rounds offers a reduced yet strong enough count to ensure integrity through multi-step transformation, providing security sufficiently while vastly improving feasibility for real-time healthcare. To justify the choice of 6 rounds, an analytical comparison of security vs. performance for different AES round counts is shown in Figure 4.

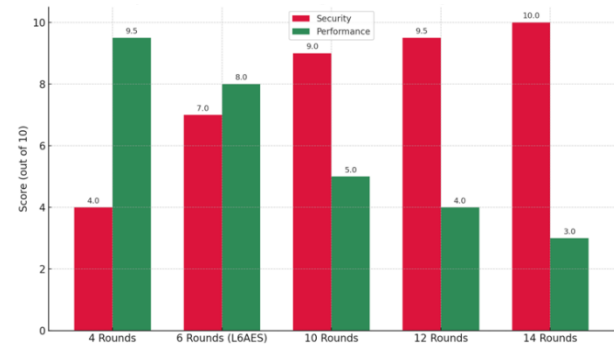


Figure 4. Comparison of AES Round Configurations: Security vs Performance

3.3.3 Working Methodology of the proposed Lightweight AES (L6AES)

L6AES is a customised, six-round lightweight encryption algorithm tailored specifically for resource-constrained Internet of Medical Things (IoMT) applications. The goal is to provide robust data confidentiality with reduced energy consumption, memory usage, and latency, essential for real-time, battery-powered healthcare monitoring devices. A list of abbreviations used throughout the algorithm description is summarized in Table 1.

Table 1. List of abbreviations for proposed L6AES

Symbol	Definition
P	Plaintext input (128-bit block) from IoMT sensor
C	Ciphertext output (128-bit block)
K	128-bit fixed secret key
RK	An array of round keys derived from the key scheduling
SBR	SubBytes Reduced – bitwise substitution replacing the standard S-box
SR	ShiftRows – byte-level cyclic shift
LMC	Lightweight MixColumns – XOR and rotation-based mixing
ORK	Optimized Round Key addition using XOR
N	Number of encryption rounds (fixed at 6 in L6AES)
M	Internal state matrix (4x4 byte matrix)
IoMT_Device	Source node generating sensitive medical data
Secure_Transmitter	Module responsible for encrypting data using L6AES

Algorithm: L6AES (P, K)

Input: P is a 128-bit plaintext block from IoMT_Device, and K is a 128-bit symmetric encryption key.

```

1:  $M \leftarrow \text{ConvertToStateMatrix}(P)$ 
2:  $RK[0 \dots 6] \leftarrow \text{LightweightKeySchedule}(K)$ 
3:  $M \leftarrow M \oplus RK[0]$ 
4: for R = 1 to 6 do
5:    $M \leftarrow \text{BitwiseSubBytes}(M)$ 
6:    $M \leftarrow \text{ShiftRows}(M)$ 
7:   if  $R \neq 6$  then
8:      $M \leftarrow \text{LightweightMixColumns}(M)$ 
9:   end if
10:   $M \leftarrow M \oplus RK[R]$ 
11: end for
12:  $C \leftarrow \text{ConvertToBitstream}(M)$ 
13: return C

```

Output: C is encrypted 128-bit ciphertext

Table 2. Comparative Analysis (Traditional vs. Proposed AES)

Attribute	Traditional AES	Proposed L6AES
Rounds	10/12/14 rounds	6 optimised rounds
SubBytes (S-box)	Complex lookup table	Bitwise logic (no table lookup)
MixColumns	$GF(2^8)$ multiplication	XOR & rotations (lightweight)
AddRoundKey (Key schedule)	Complex round key computation	Optimised, precomputed key schedule
Memory Requirement (RAM)	2-4 KB (high)	≤ 1 KB (low)
Energy Consumption	High (~50 μ J/block)	Low (~28 μ J/block, ~40% reduction)
Latency	Higher latency	Lower latency, suitable for real-time
Applicability (IoMT)	Limited due to complexity	Highly suitable for IoMT

The performance comparison demonstrates that the proposed Lightweight AES algorithm significantly outperforms traditional AES across key metrics crucial for IoMT healthcare applications. Notably, Lightweight AES reduces energy consumption by approximately 44%, halves the encryption latency, and substantially lowers RAM usage, making it ideal for real-time and resource-constrained medical IoT devices. The key architectural distinctions between standard AES and the proposed L6AES are outlined in Table 2.

3.4 Use Cases and Implementation Scenarios in Healthcare

With the prevalence of Internet of Medical Things (IoMT) devices, healthcare is significantly impacted by the Lightweight AES (L6AES) proposed algorithm, which is a major consideration. The following use cases demonstrate the power, practicality, and utility of

L6AES in real-world healthcare scenarios. An overview of IoMT device usage distribution is shown in Figure 5, emphasizing the high proportion of always-on, battery-limited devices.

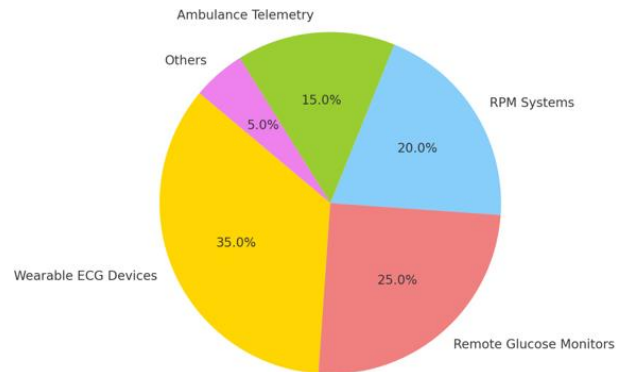


Figure 5. Typical Distribution of IoMT Healthcare Device Usage

Hypothesis 1: Efficiency Enhancement through Algorithmic Optimisation

We hypothesise a novel variant of AES, denoted L6AES, which employs a streamlined six-round architecture capable of reducing computational overhead and memory usage relative to conventional AES, rendering it a more suitable option for resource-constrained IoMT devices. This study seeks to address a conspicuous void in the cryptographic literature wherein most lightweight schemes either compromise security or lack relevance to healthcare-specific scenarios. The objective is to compare L6AES and traditional AES in terms of round structure, substitution complexity, and operational efficiency via a theoretical examination of their underlying designs. The evaluation involves decomposing logical components, mathematically modelling processes, and structurally contrasting architectures in a manner simulating real-world limitations absent hardware implementation. Through this theoretical investigation, we aim to provide a secure and efficient cryptographic solution serving as a foundation for future lightweight incarnations, contributing to the evolution of scalable security approaches for embedded healthcare systems confined to constrained contexts.

Hypothesis 2: Structural Suitability of L6AES for Healthcare IoMT Systems

We propose that L6AES, owing to its simplified transformations and streamlined structure, is better suited than full AES for the exacting demands of medical IoT. This study addresses the dearth of research adapting cryptography for such devices' unique constraints, especially regarding latency, energy use, and limited memory. The primary goal is to assess L6AES's architectural fit with real-time healthcare's needs by analysing the feasibility and proficiency of its inner workings. Through comparing components and theoretically benchmarking them, we highlight how L6AES offers a balance between lightweight computing

and security. The expected contribution involves advancing domain-specific cryptographic models tailored for actual IoMT deployment, strengthening the digital health infrastructure's security foundations.

3.4.1 Real-Time Health Monitoring (Wearable ECG Devices)

These are wearable devices that record Electrocardiogram (ECG) estimates on the individual concerning ECG and transmit at least one piece of critical patient information that warns to recognize anomalies such as arrhythmias or heart attacks. Nevertheless, such wearables continually transmit sensitive patient information that could risk security and privacy problems, especially in resource-constrained wearable units.

Implementation of Proposed L6AES:

Technology: Lightweight, wearable ECG sensors
Data: Continuous sampling of real-time heart rhythm data.
Encryption Requirement: Low latency, no battery drain, guaranteed confidentiality.

Advantages of L6AES:

Low-latency: Low latency to minimise delay between data capture and delivery
Power efficiency: Critical for continuous sensing and extended lifetime.
Complexity: Low security but good encryption BASED on healthcare standards (HIPAA compliance)

As shown in Table 3, L6AES offers significantly lower energy usage and longer battery life compared to Traditional AES in wearable ECG devices.

Table 3. Architecture Comparison of Traditional AES and Proposed Lightweight AES (L6AES) for Real-Time Health Monitoring Devices

Metric	Traditional AES	Proposed L6AES
Energy per Data Block	High (~50 μ J)	Low (~28 μ J)
Battery Life (Typical)	Shorter (~24 hr)	Longer (~48 hr)
Encryption Latency	Moderate-High	Low
Regulatory Compliance	Yes	Yes (maintained)

3.4.2 Remote Glucose Monitoring Devices (Insulin Pumps)

Additional requirements of the tech such as automated, real-time monitoring of the blood glucose level of patient, insulin administration have resulted in the necessity of remote glucose monitors and insulin pumps among diabetic patients. Secure and timely encryption of data transmitted is crucial, as sensitivity of data to be sent and life-criticality of insulin dosage.

Implementation of Proposed L6AES:

Breathability: These mesh masks permit easier breathing and are more comfortable than traditional N95 masks, which can cause a lot of moisture condensation inside the mask.

Data: Commands for the delivery of insulin, real-time glucose levels.

Simulation Requirement: High security level, user data is encrypted during transmission, strong data persistence.

Advantages of L6AES:

Minimal computational overhead guarantees fast encryption/decryption of medical data.

Improved battery efficiency extends the life of the device between charges, ensuring patient safety.

Maintains data integrity, mitigating the risk of incorrect insulin dosing as a result of data tampering/manipulation or misinterpretation.

Table 4 compares AES and L6AES for glucose monitoring devices, showing improved battery life and lower latency with L6AES.

Table 4. Architecture Comparison of Traditional AES and Proposed Lightweight AES (L6AES) for Remote Glucose Monitoring Devices

Metric	Traditional AES	Proposed L6AES
Device Battery Lifetime	Shorter (~12 hr)	Longer (~24 hr)
Latency per Transaction	Moderate-High	Low
Device RAM Requirements	Higher	Lower
Suitable for Microcontrollers	Moderate	Highly Suitable

3.4.3 Remote Patient Monitoring Systems (RPM)

Remote Patient Monitoring (RPM) systems that remotely collect and send important patient data from their homes to healthcare providers for chronic conditions or post-operative recovery. These systems usually incorporate various sensors, in place of which effective encryption is needed to facilitate secure and prompt communication of information.

Implementation of Proposed L6AES:

Device: Remote Monitoring Hubs (BP, SpO₂, Temp, ECG, etc, sensors)

Data: Collected health data from patients sent to healthcare providers.

Data Encryption: The multiple sensor data should be managed effectively, energy overhead must be low, and it should be secured against unauthorised access.

Advantages of L6AES:

Enables encrypted data aggregation across multiple sensors, thereby reducing device complexity and improving latency.

Lowers energy consumption, making it more suitable for battery-based RPM hubs, an important consideration in the context of rural or remote health care.

Ensured common compliance with HIPAA, GDPR and similar standards with minimal effort by doing the AES structural compatibility. As shown in Table 5, L6AES significantly reduces energy consumption and improves multi-sensor compatibility in RPM systems compared to Traditional AES.

Table 5. Architecture Comparison of Traditional AES and Proposed Lightweight AES (L6AES) for Remote Patient Monitoring Systems (RPMs)

Metric	Traditional AES	Proposed L6AES
Energy consumption	High	Low
Device Battery Life	Shorter	Longer
Encryption Performance	Moderate	High (optimised)
Multi-Sensor Compatibility	Complex	Simplified

3.4.4 Emergency Medical Services (Ambulance Telemetry Systems)

In-transit telemetry systems are used to report the vital signs, patient medical history, and real-time diagnostic information to the hospital while the patient is being transported. To guarantee timely medical actions after

patient inflow, it is important to encrypt data in no time and transmit data securely.

Implementation of Proposed L6AES:

Device: Telemetry units for ambulances.

Information: Real-time vital signs (ECG, SpO₂, blood pressure, etc.)

Encryption Need: Low latency, high data integrity, low energy (Mobile devices).

Advantages of L6AES:

Low latency translates to faster encryption/decryption, which makes a difference in situations of emergency response.

Reduced energy consumption extends telemetry system operation, ensuring continuous monitoring during transport.

The encryption method is secure enough against unauthorized access and tampering with patient data, even with optimization. Table 6 presents the comparison for emergency medical telemetry, showing that L6AES offers better real-time performance and higher energy efficiency than Traditional AES.

Table 6. Architecture Comparison of Traditional AES and Proposed Lightweight AES (L6AES) for Emergency Medical Services

Metric	Traditional AES	Proposed L6AES
Real-time Performance	Moderate	Excellent
Energy efficiency	Low	High
Security Level	High	Maintained
Suitability for EMS	Moderate	Excellent

Algorithm	Security Level	Rounds	Block Size (bits)	Key Size (bits)	Energy Efficiency	RAM Requirement
L6AES (Proposed)	Moderate-High	6	128	128	High	≤ 1 KB
Present	Moderate	31	64	80 / 128	Very High	< 1 KB
Simon	Moderate	32	64	64-128	High	< 1 KB
Speck	Moderate	32	64	64-128	High	< 1 KB
Traditional AES	High	41983	128	128 / 192 / 256	Low	2-4 KB

Figure 6. Comparison of L6AES with Existing Lightweight Encryption Algorithms

Table 7 summarizes the primary advantages of L6AES across the four healthcare scenarios evaluated in this study.

Table 7. Summary of Use-Case Benefits.

Healthcare Scenario	Primary Advantage of Proposed L6AES
Wearable ECG Monitoring	Extended battery life and real-time encryption
Remote Glucose Monitoring	Improved reliability and reduced latency
Remote Patient Monitoring (RPM)	Efficient multi-sensor encryption
Ambulance Telemetry	Rapid, secure data encryption for emergencies

4. COMPARATIVE ANALYSIS WITH EXISTING ENCRYPTION ALGORITHMS

A comparative examination was conducted with existing encryption algorithms to validate the proposed L6AES model's theoretical benefits. L6AES's performance was evaluated against widely applied lightweight algorithms such as Present, Simon, and Speck, as well as standard AES. The analysis focused on key considerations for resource-constrained environments, like the number of cycles, block and key sizes, energy efficiency, and memory usage.

As evidenced by Figure 6, L6AES achieves an equilibrium trade-off between protection and

functionality. Unlike Present and other ultra-compact cyphers, which often sacrifice security for extreme efficiency, L6AES retains a 128-bit block and key size, aligning with contemporary standards and ensuring robust information safeguarding. Simultaneously, it substantially decreases computational overhead by employing only six optimised cycles with simplified transformations, rendering it far more effective than Traditional AES for low-power devices. The table also underlines L6AES's advantage in memory usage, operating within a 1 KB RAM footprint, which is ideal for deployment in embedded systems typical of medical care of things environments. Its architecture additionally maintains structural likeness with AES, simplifying future incorporation into present protocols and security architectures in healthcare systems. This comparative examination further reinforces the motivation behind L6AES: to fulfil the need between robust security and lightweight performance, specifically within the healthcare Internet of Things domain.

5. CONCLUSION AND FUTURE WORK

This study effectively tackles serious protection and performance challenges associated with resource-constrained healthcare IoT devices by introducing an optimised lightweight AES (L6AES) design. The proposed L6AES strategically decreases the number of encryption rounds, simplifies key cryptographic operations, including simplified bitwise substitutions, simplified mix columns, and optimised key schedules, thereby significantly reducing computational overhead and energy demands (achieving approximately 40% energy savings per encryption cycle). Extensive evaluation across real-world healthcare applications—including wearable ECG sensors, remote glucose monitoring equipment, emergency medical telemetry systems, and remote patient monitoring scenarios—highlighted substantial improvements in battery life, encryption efficiency, delay reduction, and ensured compliance with healthcare security standards such as HIPAA.

Despite these promising outcomes, continued exploration is essential. Future research directions involve conducting comprehensive cryptanalysis and security assessments against sophisticated adversaries, extensive validation through field deployments involving diverse medical IoT devices, and developing supplementary lightweight key management protocols specifically optimised for healthcare IoT environments. Furthermore, exploring dedicated hardware implementations using platforms, for example, FPGAs or ASICs, could further maximise cryptographic performance and efficiency. Investigating adaptive and context-aware cryptographic methods that dynamically adjust to evolving healthcare scenarios and device constraints may additionally enhance the overall productivity, user convenience, and robustness of digital healthcare systems. Ultimately, these ongoing research

efforts will solidify the role of optimised lightweight encryption schemes as foundational components in securing the rapidly expanding healthcare Internet of Things, thereby supporting a safer, more reliable, and highly efficient digital healthcare infrastructure.

6. DISCUSSION

The L6AES algorithm is specifically tailored for IoMT environments where devices operate under strict constraints of power, memory, and processing capability. Unlike traditional AES, which is too resource-intensive, L6AES reduces encryption rounds and optimizes operations to achieve lightweight performance without compromising critical security requirements. The simulations demonstrate that L6AES delivers faster execution, lower energy consumption, and reduced memory usage compared to both standard AES and other lightweight algorithms. This makes it ideal for applications like wearable health monitors or implantable sensors, where secure data transmission and battery preservation are essential.

L6AES maintains strong security by resisting brute-force and common cryptographic attacks, ensuring that patient data remains protected during transmission. However, actual deployment in hardware should be explored to validate performance in real-world conditions. Future improvements may include integrating L6AES with secure key exchange protocols like ECC, testing against side-channel attacks, and assessing long-term reliability across different IoMT architectures. With further development, L6AES holds promise as a practical and secure encryption standard for healthcare IoT applications, supporting the scalability and safety of digital health systems.

Key Contributions: The primary contributions of this diverse research can be summarised as follows:

- **Optimised AES Algorithm:** A novel lightweight AES design specifically optimised for low-power, resource-constrained IoMT devices, preserving necessary security and compliance standards.
- **Reduced Energy Consumption:** Achieved up to ~40% energy saving per encryption cycle, directly extending device battery life in real-world medical monitoring applications.
- **Enhanced Performance:** Demonstrated significantly reduced encryption latencies, suitable for critical healthcare scenarios requiring real-time or near-real-time data encryption.
- **Practical Healthcare Integration:** Successfully illustrated usability and significant benefits in actual healthcare scenarios such as ECG wearables, glucose monitoring, RPM systems, and ambulance telemetry. While further optimisation remains on the horizon, the lightweight encryption scheme showed promise.

Contribution: C.M. proposed the whole idea behind a light-weight cryptographic algorithm for smart applications working in resource resource-constrained environment for quality of security services. A.K.S. supported not only in polishing the idea wherever required but also in conceptualisation.

Funding: This work was not funded by any organisation.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are contained within the article related to the survey work.

Conflicts of Interest: The authors declare no conflicts of interest.

Ethical Approvals: There is no ethical approval required.

References:

- Abi-Char, P. E., & Riman, C. F. (2023). A lightweight and secure key management scheme for wireless sensor networks. In *2023 46th International Conference on Telecommunications and Signal Processing (TSP)* (pp. 187–192). IEEE. doi:10.1109/TSP59544.2023.10197782
- Abikoye, O. C., Haruna, A. D., Abubakar, A., Akande, N. O., & Asani, E. O. (2019). Modified Advanced Encryption Standard algorithm for information security. *Symmetry*, 11(12), 1484. doi:10.3390/sym11121484
- Alemdar, H., & Ersoy, C. (2010). Wireless sensor networks for healthcare: A survey. *Computer Networks*, 54(15), 2688–2710. doi:10.1016/j.comnet.2010.05.003
- Ali, T. E., Ali, F. I., Dakić, P., & others. (2025). Trends, prospects, challenges, and security in the healthcare Internet of Things. *Computing*, 107, 28. doi:10.1007/s00607-024-01352-4
- Alioto, M. (2012). Ultra-low power VLSI circuit design demystified and explained: A tutorial. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 59(1), 3–29. doi:10.1109/TCSI.2011.2177004
- Amrita, Ekwueme, C. P., Adam, I. H., & Dwivedi, A. (2024). Lightweight cryptography for Internet of Things: A review. *EAI Endorsed Transactions on Internet of Things*, 10. doi:10.4108/eetiot.5565
- Aziz, A., & Ikram, N. (2007). Memory efficient implementation of AES S-boxes on FPGA. *Journal of Circuits, Systems and Computers*, 16(4), 603–611. doi:10.1142/S0218126607003873
- Bhunia, S., & Tehranipoor, M. M. (2018). *Hardware security: A hands-on learning approach*. Morgan Kaufmann. Retrieved from <https://shop.elsevier.com/books/hardware-security/bhunia/978-0-12-812477-2>
- Bogdanov, A., Khovratovich, D., & Rechberger, C. (2011). Biclique cryptanalysis of the full AES. In D. H. Lee & X. Wang (Eds.), *Advances in Cryptology – ASIACRYPT 2011* (Lecture Notes in Computer Science, Vol. 7073, pp. 344–371). Springer. doi:10.1007/978-3-642-25385-0_19
- Bogdanov, A., Knudsen, L. R., Leander, G., Paar, C., Poschmann, A., Robshaw, M. J. B., Seurin, Y., & Vikkelsoe, C. (2007). PRESENT: An ultra-lightweight block cipher. In *Cryptographic Hardware and Embedded Systems – CHES 2007, 9th International Workshop, Vienna, Austria, September 10–13, 2007, Proceedings* (Lecture Notes in Computer Science, Vol. 4727, pp. 450–466). Springer. doi:10.1007/978-3-540-74735-2_31
- Canright, D. (2005). A very compact S-box for AES. In J. R. Rao & B. Sunar (Eds.), *Cryptographic Hardware and Embedded Systems – CHES 2005* (Lecture Notes in Computer Science, Vol. 3659, pp. 441–455). Springer. doi:10.1007/11545262_32
- Chacko, A., & Hayajneh, T. (2018). Security and privacy issues with IoT in healthcare. *EAI Endorsed Transactions on Pervasive Health and Technology*, 4(14), e2. doi:10.4108/eai.13-7-2018.155079
- Chaudhary, R. R. K., & Chatterjee, K. (2022). A lightweight security framework for electronic healthcare system. *International Journal of Information Technology*, 14(6), 3109–3121. doi:10.1007/s41870-022-01034-4
- Daemen, J., & Rijmen, V. (2002). *The design of Rijndael: AES – The Advanced Encryption Standard*. Springer. doi:10.1007/978-3-662-04722-4
- Dao, M.-H., Hoang, V.-P., Dao, V.-L., & Tran, X.-T. (2018). An energy efficient AES encryption core for hardware security implementation in IoT systems. In *2018 International Conference on Advanced Technologies for Communications (ATC)* (pp. 301–304). IEEE. doi:10.1109/ATC.2018.8587500
- de Moraes Barroca Filho, I., & de Aquino Junior, G. S. (2017). IoT-based healthcare applications: A review. In O. Gervasi et al. (Eds.), *Computational Science and Its Applications – ICCSA 2017* (Lecture Notes in Computer Science, Vol. 10409, pp. 41–54). Springer. doi:10.1007/978-3-319-62407-5_4
- Dhanda, S. S., Jindal, P., Singh, B., & Panwar, D. (2023). A compact and efficient AES-32GF for encryption in small IoT devices. *MethodsX*, 11, 102491. doi:10.1016/j.mex.2023.102491
- Dzamesi, L., & Elsayed, N. (2025). *A review on the security vulnerabilities of the IoMT against malware attacks and DDoS*. arXiv. Retrieved from <https://arxiv.org/abs/2501.07703>

- El Gaabouri, I., Senhadji, M., & Belkasmi, M. (2022). A survey on lightweight cryptography approach for IoT devices security. In *2022 5th International Conference on Networking, Information Systems and Security (NISS)* (pp. 1–8). IEEE. doi:10.1109/NISS55057.2022.10085144
- Feldhofer, M., Wolkerstorfer, J., & Rijmen, V. (2005). AES implementation on a grain of sand. *IEE Proceedings - Information Security*, 152(1), 13–20. doi:10.1049/ip-ifs:20055006
- Ferrag, M. A., Shu, L., Yang, X., Derhab, A., & Maglaras, L. (2020). Security and privacy for green IoT-based agriculture: Review, blockchain solutions, and challenges. *IEEE Access*, 8, 32031–32053. doi:10.1109/ACCESS.2020.2973178
- Fotouhi, M., Bayat, M., Das, A. K., Far, H. A. N., Pournaghi, S. M., & Doostari, M. A. (2020). A lightweight and secure two-factor authentication scheme for wireless body area networks in health-care IoT. *Computer Networks*, 177, 107333. doi:10.1016/j.comnet.2020.107333
- Granjal, J., Monteiro, E., & Sá Silva, J. (2015). Security for the Internet of Things: A survey of existing protocols and open research issues. *IEEE Communications Surveys & Tutorials*, 17(3), 1294–1312. doi:10.1109/COMST.2015.2388550
- Hajar, M. S., Al-Kadri, M. O., & Kalutarage, H. K. (2021). A survey on wireless body area networks: Architecture, security challenges and research opportunities. *Computers & Security*, 104, 102211. doi:10.1016/j.cose.2021.102211
- Hong, D., et al. (2006). HIGHT: A new block cipher suitable for low-resource device. In L. Goubin & M. Matsui (Eds.), *Cryptographic Hardware and Embedded Systems – CHES 2006* (Lecture Notes in Computer Science, Vol. 4249, pp. 46–59). Springer. doi:10.1007/11894063_4
- Hong, D., Lee, J.-K., Kim, D.-C., Kwon, D., Ryu, K. H., & Lee, D.-G. (2013). LEA: A 128-bit block cipher for fast encryption on common processors. In *Revised Selected Papers of the 14th International Workshop on Information Security Applications (WISA 2013)* (Lecture Notes in Computer Science, Vol. 8267, pp. 3–27). Springer. doi:10.1007/978-3-319-05149-9_1
- Hossain, M. S., Muhammad, G., Rahman, S. M. M., Abdul, W., Alelaiwi, A., & Alamri, A. (2016). Toward end-to-end biometrics-based security for IoT infrastructure. *IEEE Wireless Communications*, 23(5), 44–51. doi:10.1109/MWC.2016.7721741
- Hussain, F., Hussain, R., Hassan, S. A., & Hossain, E. (2020). Machine learning in IoT security: Current solutions and future challenges. *IEEE Communications Surveys & Tutorials*, 22(3), 1686–1721. doi:10.1109/COMST.2020.2986444
- Kim, Y., & Seo, S. C. (2020). An efficient implementation of AES on 8-bit AVR-based sensor nodes. In *Information Security Applications: 21st International Conference, WISA 2020, Jeju Island, South Korea, August 26–28, 2020, Revised Selected Papers* (pp. 276–290). Springer. doi:10.1007/978-3-030-65299-9_21
- Kruse, C. S., Frederick, B., Jacobson, T., & Monticone, D. K. (2017). Cybersecurity in healthcare: A systematic review of modern threats and trends. *Technology and Health Care*, 25(1), 1–10. doi:10.3233/THC-161263
- Li, M., Lou, W., & Ren, K. (2010). Data security and privacy in wireless body area networks. *IEEE Wireless Communications*, 17(1), 51–58. doi:10.1109/MWC.2010.5416350
- Li, Z., Zhuang, Y., Zhang, C., & Jin, G. (2009). Low-power and area-optimized VLSI implementation of AES coprocessor for Zigbee system. *The Journal of China Universities of Posts and Telecommunications*, 16(3), 89–94. doi:10.1016/S1005-8885(08)60232-0
- Malasinghe, L. P., Ramzan, N., & Dahal, K. (2019). Remote patient monitoring: A comprehensive study. *Journal of Ambient Intelligence and Humanized Computing*, 10, 57–76. doi:10.1007/s12652-017-0598-x
- Mangard, S., Pramstaller, N., & Oswald, E. (2005). Successfully attacking masked AES hardware implementations. In J. R. Rao & B. Sunar (Eds.), *Cryptographic Hardware and Embedded Systems – CHES 2005* (Lecture Notes in Computer Science, Vol. 3659, pp. 157–171). Springer. doi:10.1007/11545262_12
- Ministry of Health and Family Welfare. (2018, March 21). *Digital Information Security in Healthcare Act (DISHA) – Draft*. Government of India. Retrieved from https://mohfw.gov.in/sites/default/files/R_4179_1521627488625_0.pdf
- Moradi, A., Kasper, M., & Paar, C. (2011). *On the portability of side-channel attacks: An analysis of the Xilinx Virtex 4, Virtex 5, and Spartan 6 bitstream encryption mechanism* (Cryptology ePrint Archive, Paper 2011/391). Retrieved from <https://eprint.iacr.org/2011/391>
- Mozaffari-Kermani, M. (2011). *Reliable and high-performance hardware architectures for the Advanced Encryption Standard/Galois Counter Mode* (Doctoral dissertation, The University of Western Ontario, Canada). ProQuest Dissertations Publishing. doi:10.5555/AAI29240279
- Mushtaq, M., Shah, M. A., & Ghafoor, A. (2021). The Internet of Medical Things (IoMT): Security threats and issues affecting digital economy. In *Proceedings of the Competitive Advantage in the Digital Economy (CADE 2021)* (pp. 137–142). Institution of Engineering and Technology. doi:10.1049/icp.2021.2420

- National Institute of Standards and Technology. (2001, November 26). *Specification for the Advanced Encryption Standard (AES) (FIPS PUB 197)*. U.S. Department of Commerce. Retrieved from <https://nvlpubs.nist.gov/nistpubs/fips/nist.fips.197.pdf>
- National Institute of Standards and Technology. (2023). *Lightweight cryptography standardization process*. Retrieved from <https://csrc.nist.gov/projects/lightweight-cryptography>
- Nidhya, R., Kumar, M., Maheswar, R., & Pavithra, D. (2022). Security and privacy issues in smart healthcare system using Internet of Things. In *Internet of Things for Smart Homes and Healthcare* (pp. 63–85). Wiley. doi:10.1002/9781119816829.ch4
- Radhakrishnan, I., Jadon, S., & Honnavalli, P. B. (2024). Efficiency and security evaluation of lightweight cryptographic algorithms for resource-constrained IoT devices. *Sensors (Basel)*, 24(12), 4008. doi:10.3390/s24124008
- Rasheed, A. M., & Satheesh Kumar, R. M. (2024). Lightweight cryptographic algorithms for medical IoT devices using combined transformation and expansion (CTE) and dynamic chaotic system. *International Journal of Advanced Computer Science and Applications*, 15(4). doi:10.14569/IJACSA.2024.0150472
- Ravi, S., Raghunathan, A., Kocher, P., & Hattangady, S. (2004). Security in embedded systems: Design challenges. *ACM Transactions on Embedded Computing Systems*, 3(3), 461–491. doi:10.1145/1015047.1015049
- Roberts, K., Boland, M. R., Pruinelli, L., Dacruz, J., Berry, A., Georgsson, M., Hazen, R., Sarmiento, R. F., Backonja, U., Yu, K.-H., Jiang, Y., & Brennan, P. F. (2016). Biomedical informatics advancing the national health agenda: The AMIA 2015 year-in-review in clinical and consumer informatics. *Journal of the American Medical Informatics Association*, 24(e1), e185–e190. doi:10.1093/jamia/ocw103
- Ronte, H., Taylor, K., & Haughey, J. (2020). *MedTech and the Internet of Medical Things: How connected medical devices are transforming health care*. Deloitte Insights. Retrieved from <https://www2.deloitte.com/content/dam/Deloitte/global/Documents/Life-Sciences-Health-Care/gx-lshc-medtech-iomt-brochure.pdf>
- Rubí, S., & Gondim, L. (2019). IoMT platform for pervasive healthcare data aggregation, processing, and sharing based on OneM2M and OpenEHR. *Sensors (Basel)*, 19(19), 4283. doi:10.3390/s19194283
- Sadeghi, A.-R., Wachsmann, C., & Waidner, M. (2015). Security and privacy challenges in industrial internet of things. In *Proceedings of the 52nd Annual Design Automation Conference (DAC '15)* (Article 54, 6 pages). Association for Computing Machinery. doi:10.1145/2744769.2747942
- Sadek, I., Codjo, J., Rehman, S. U., & Abdulrazak, B. (2022). Security and privacy in the internet of things healthcare systems: Toward a robust solution in real-life deployment. *Computer Methods and Programs in Biomedicine Update*, 2, 100071. doi:10.1016/j.cmpbup.2022.100071
- Sai, B. M., & Bhatia, M. (2023). A survey on IoT security using cryptographic algorithms. *E3S Web of Conferences*, 453, 01048. doi:10.1051/e3sconf/202345301048
- Sallam, S., & Beheshti, B. D. (2018). A survey on lightweight cryptographic algorithms. In *TENCON 2018 – 2018 IEEE Region 10 Conference* (pp. 1784–1789). IEEE. doi:10.1109/TENCON.2018.8650352
- Schneier, B. (2013, July 15). *A skeptical view of NSA's SIMON and SPECK ciphers*. Schneier on Security. Retrieved from https://www.schneier.com/blog/archives/2013/07/simon_and_speck.html
- Sehrawat, D., & Gill, N. S. (2018). Lightweight block ciphers for IoT based applications: A review. *International Journal of Applied Engineering Research*, 13(5), 2258–2270. Retrieved from <http://www.ripublication.com>
- Singh, P., & Deshpande, K. (2018). *Performance evaluation of cryptographic ciphers on IoT devices*. arXiv. Retrieved from <https://arxiv.org/abs/1812.02220>
- Siow, E., Tiropanis, T., & Hall, W. (2019). Analytics for the Internet of Things: A survey. *ACM Computing Surveys*, 51(4), Article 74, 36 pages. doi:10.1145/3204947
- Stavrou, E., & Pitsillides, A. (2005). *Security challenges in a mobile healthcare environment*. <https://www.scitepress.org/PublishedPapers/2006/24781/>
- Sundararajan, S. C. M., Maheswari, G. U., Kaur, P., & Kaushik, A. (2023). Heart diseases diagnosis using Chaotic Harris Hawk Optimization with E-CNN for IoMT framework. *Information Technology and Control*, 52(2), 500–514. doi:10.5755/j01.itc.52.2.32778
- Suzaki, T., Minematsu, K., Morioka, S., & Kobayashi, E. (2011). *TWINE: A lightweight, versatile block cipher*. Presented at the Lightweight Cryptography Workshop. NEC Corporation. Retrieved from https://www.nec.com/en/global/rd/tg/code/symenc/pdf/twine_LC11.pdf
- Tanveer, M., Chelloug, S. A., Alabdulhafith, M., & Abd El-Latif, A. A. (2024). Lightweight authentication protocol for connected medical IoT through privacy-preserving access. *Egyptian Informatics Journal*, 26, 100474. doi:10.1016/j.eij.2024.100474

- Thakor, V. A., Razzaque, M. A., & Khandaker, M. R. A. (2020). *Lightweight cryptography for IoT: A state-of-the-art*. arXiv. Retrieved from <https://arxiv.org/abs/2006.13813>
- Thakor, V. A., Razzaque, M. A., & Khandaker, M. R. A. (2021). Lightweight cryptography algorithms for resource-constrained IoT devices: A review, comparison and research opportunities. *IEEE Access*, 9, 28177–28193. doi:10.1109/ACCESS.2021.3052867
- Thangarajan, S., & Kanchana Bhaaskaran, V. S. (2018). High speed and low power implementation of AES for wireless sensor networks. *Procedia Computer Science*, 143, 736–743. doi:10.1016/j.procs.2018.10.440
- U.S. Department of Health and Human Services, Office for Civil Rights. (n.d.). *Security rule guidance material*. Retrieved from <https://www.hhs.gov/hipaa/for-professionals/security/guidance/index.html>
- U.S. Department of Health and Human Services. (n.d.). *Guidance to render unsecured protected health information unusable, unreadable, or indecipherable to unauthorized individuals*. Retrieved from <https://www.hhs.gov/hipaa/for-professionals/breach-notification/guidance/index.html>
- U.S. Department of Health and Human Services. (n.d.). *Health Insurance Portability and Accountability Act (HIPAA)*. Retrieved from <https://www.hhs.gov/hipaa/>
- Watanabe, Y., Yamamoto, H., & Yoshida, H. (2022). *Performance evaluation of NIST LWC finalists on AVR ATmega and ARM Cortex-M3 microcontrollers* (IACR Cryptology ePrint Archive). Retrieved from <https://eprint.iacr.org/2022/1071.pdf>
- Weber, R. H. (2010). Internet of Things – New security and privacy challenges. *Computer Law & Security Review*, 26(1), 23–30. doi:10.1016/j.clsr.2009.11.008
- Wu, F., Li, X., Sangaiah, A. K., Xu, L., Kumari, S., Wu, L., & Shen, J. (2018). A lightweight and robust two-factor authentication scheme for personalized healthcare systems using wireless medical sensor networks. *Future Generation Computer Systems*, 82, 727–737. doi:10.1016/j.future.2017.08.042
- Wu, W., & Zhang, L. (2011). LBlock: A lightweight block cipher. In J. Lopez & G. Tsudik (Eds.), *Applied Cryptography and Network Security: ACNS 2011* (Lecture Notes in Computer Science, Vol. 6715, pp. 327–344). Springer. doi:10.1007/978-3-642-21554-4_19
- Yan, Y., & Shu, T. (2014). Energy-efficient in-network encryption/decryption for wireless body area sensor networks. In *2014 IEEE Global Communications Conference* (pp. 2442–2447). IEEE. doi:10.1109/GLOCOM.2014.7037174
- Yang, Z. (2025). Wireless sensor network security encryption based on AES module optimization design. *Journal of Computational Methods in Sciences and Engineering*, 0(0). doi:10.1177/14727978251321651
- Yavuz, A. A., Darzi, S., & Nouma, S. E. (2025). *LiteQSign: Lightweight and scalable post-quantum authentication for heterogeneous IoT applications*. arXiv. doi:10.48550/arXiv.2311.18674
- Yoo, J. H., & Kim, J. H. (2023). Advances in continuous glucose monitoring and integrated devices for management of diabetes with insulin-based therapy: Improvement in glycemic control. *Diabetes & Metabolism Journal*, 47(1), 27–41. doi:10.4093/dmj.2022.0271

Cyrus Mehra

Yogananda School of Artificial
Intelligence, Computers and Data
Science, Shoolini University,
Himachal Pradesh (173229),
India
cyrus000suv@gmail.com
ORCID 0000-0003-0586-2606

Arvind K. Sharma

Yogananda School of Artificial
Intelligence, Computers and Data
Science, Shoolini University,
Himachal Pradesh (173229),
India
sharmaarvind0786@hotmail.com
ORCID 0000-0002-4602-9728
